

SCIENTIFIC CLAIM AUDIT / REVIEWER EDITION

Find the weakness in your paper before reviewers do



An academic paper can look convincing while hiding problems across its manuscript, data, code, supplementary materials, or preregistration. Finding those problems after submission can mean major revisions, rejection, or months of avoidable work. ClaimBounty gives you an independent scientific audit before the paper reaches reviewers.

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AUTHORSHIP**Authored by Matheus Paranhos**

Public hackathon repository projection

DRAFT

CLAIMBOUNTY ATTEMPT UNSCORED; CLEAN-CLONE TEST AND RELEASE REVIEW REMAIN OPEN

FIVE-MINUTE REVIEWER PATH

A fast review without blurring the boundaries.

The website collects and prepares evidence. The scientific audit runs locally with Toone. The current evidence includes a scored invalid comparator and an unscored ClaimBounty protocol deviation, not a qualified same-case comparison.

00:00 Read the product boundary

ClaimBounty serves researchers, authors, reviewers, and research teams who need an evidence-linked check of a quantitative claim before submission or decision.

01:00 Open the interface or run the visual preview

Review intake, email verification, and administrative surfaces. A visual preview does not start the API, storage, scanner, database, mail sandbox, or worker.

02:00 Inspect the local workflow package

Confirm the parent workflow, six child workflows, project template, local runner, and integrity manifests.

03:00 Read evidence with its classification

Separate observed runtime from estimated human effort, token usage from monetary cost, and the terminal ChatGPT score from the unscored ClaimBounty protocol deviations.

04:00 Run the public package checks

```
node scripts/generate-public-manifests.mjs --check
scripts/check-public-links.mjs
scripts/check-public-release
```

Detailed human-estimate basis: docs/HUMAN_BASELINE.md, including eight work packages, assumptions, exclusions, sensitivity drivers, and cited sources. Machine record: submission/evidence/human-estimate.json.

The comparator was adjudicated 1/100 and invalid incomplete. The ClaimBounty attempt is unscored, so do not derive a head-to-head comparison, speedup, or dollar cost from this draft.

CHOOSE THE RIGHT PATH

Four modes, four different claims.

MODE	PREREQUISITES	EXPECTED RESULT	DOES NOT PROVE
Visual preview	Node 22 from <code>.node-version</code> ; pnpm 9.15.0	Static interface at the Vite preview URL, normally <code>http://localhost:4173</code>	API behavior, upload, email, storage, scanning, worker, or export
Hosted stack	Git, Docker Engine, Docker Compose v2, free local ports, local <code>.env</code>	Web at <code>http://127.0.0.1:8081</code> ; mail at <code>http://localhost:8025</code>	Scientific execution, adjudication, payment, or publication
Scientific workflow	macOS, latest native <u>Toone</u> , one supported coding-agent client (Codex or Claude Code), Python 3.11+, study runtimes, verified case	Local audit artifacts under project-bound paths and a reviewer package	Release verification of the Explore Workflows listing or benchmark qualification
Contributor gates	Hosted prerequisites plus Go 1.26.6; more disk and time for full gates	Contract, architecture, unit, browser, system, package, and release checks as selected	Production readiness without secrets, TLS, identity, lifecycle, monitoring, and policy review

Visual-only preview

```
pnpm install --frozen-lockfile
pnpm --filter @micro1/web build
pnpm --filter @micro1/web preview
```

Expected terminal output includes the local Vite URL. API-backed actions will not complete in this mode.

Hosted stack

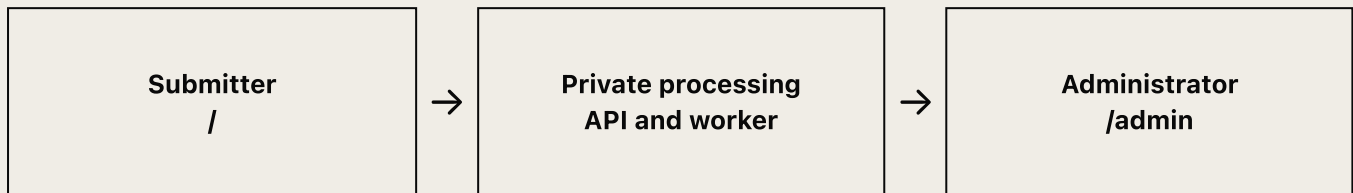
```
cp .env.example .env
docker compose --project-name claimbounty-
public \
  --file infra/compose.yaml --profile
claimbounty \
  up --build -d --wait
```

Replace demonstration credentials before any exposure beyond the local machine. Use synthetic data for public demonstrations.

Stop without deleting volumes: `docker compose --project-name claimbounty-public --file infra/compose.yaml --profile claimbounty down`

SUBMITTER AND ADMINISTRATOR PATH

From private upload to authorized export.



SUBMITTER

1. Select paper and materials

Open <http://127.0.0.1:8081>. Add exactly one PDF marked **Primary paper**, then optional documents, data, code, notebooks, or a ZIP archive. The interface accepts up to 20 files, 250 MB each, and 1 GB total. The primary PDF is limited to 50 MB.

SUBMITTER

2. Verify email

Request the six-digit code, open the local mail sandbox at <http://localhost:8025>, and enter the code within ten minutes. The session is held in a secure browser cookie and not browser storage.

SUBMITTER

3. Define and authorize

Enter the study title, review purpose, exact target claim, and claim location. Record code-execution, external-search, participant-data, and direct-identifier choices. Uploads and analysis use require explicit authorization before submission.

SYSTEM

4. Inspect privately

Files enter private malware inspection. The receipt exposes a private reference and file count. The website does not open research code or dispatch a scientific audit.

ADMINISTRATOR

5. Review and freeze intake

Sign in at <http://127.0.0.1:8081/admin> with the configured allowlisted email. Review metadata, privacy declarations, file scan states, and the server-owned retention policy. Freeze the local handoff intake.

ADMINISTRATOR

6. Create and download export

When every included file is clean, readiness checks pass, and the trusted routine revision is current, create the immutable export. Download the ZIP and retain the displayed SHA-256 and RFC 9530 Content-Digest receipt.

Local demonstration email is not evidence of production deliverability. The default administrator is the configured local administrator address; it is a local fixture, not a production credential.

DATA BOUNDARY

The transfer is an artifact, not an execution bridge.

HOSTED BOUNDARY**Private intake and policy**

1. Browser and web application
2. API and PostgreSQL state
3. Private object storage
4. Scanner and worker
5. Administrative authorization

LOCAL BOUNDARY**Scientific execution and review**

1. Whole-archive digest verification
2. Local Toone workflow
3. Study-specific runtimes
4. Independent verification
5. Audit package assembly

AUTHORIZED IMMUTABLE ZIP + TRUSTED SHA-256 + CONTENT-DIGEST → VERIFY BEFORE PARSING

Equivalent of the repository architecture diagram. Hosted credentials cannot dispatch routines. Local execution requires a separate operator decision after verification.

Hosted protections

- Server-side owner and administrator authorization on every resource read.
- Private buckets and random object keys with no email or filename.
- Bounded upload sizes, file validation, malware scanning, and fail-closed export readiness.
- Frozen source and personal-data deletion deadlines that administrators may shorten, not extend.

Public projection exclusions

- Customer data and private submissions.
- Hidden answer keys and restricted screenshots.
- Internal organization configuration and absolute workstation paths.
- The prior assessment bundle and benchmark files not cleared for redistribution.

Offline archive verification

```
make verify-export EXPORT_SHA256=<64-lowercase-hex>
```

Place the archive at `exports/claimbounty-export.zip`. The verifier checks the expected whole-archive digest before ZIP parsing and extracts only into a new verified destination.

NATIVE MACOS PATH

Install, import, select, run.

Scientific execution is local. Use the latest Toone release on macOS with one coding-agent client: Codex or Claude Code. The workflow export installs the parent and child workflows, schemas, templates, and local runner.

- ☐ **Install the latest Toone release.** Download the native macOS application from trytoone.com.
- ☐ **Install one supported coding-agent client.** Choose Codex or Claude Code, then complete that client's official sign-in flow. The next page gives the two official setup paths. Keep provider credentials outside the repository.
- ☐ **Create a project.** Open Toone, create a new project, and select an empty directory for it.
- ☐ **Add the research files.** Create `claimbounty/input/case-bundle/` in the project directory. Put the manuscript, data, code, supplements, preregistration, environment files, and authorized supporting sources there.
- ☐ **Import ClaimBounty.** Open *Explore Workflows*, select `micro1/ClaimBounty`, and import it. The release export should install every packaged workflow and resource.
- ☐ **Bind the case.** Select `claimbounty/input/case-bundle/` for the `case-bundle` input, supply the remaining dispatch documents, and confirm every study-specific runtime.
- ☐ **Run the parent workflow.** Review the bound inputs, start ClaimBounty, and treat all supplied research code as untrusted. If the release listing is unavailable, follow the repository fallback in `docs/REPRODUCE.md`.

Package verification

```
node scripts/generate-public-manifests.mjs
--check
(cd workflow/claimbounty-scientific-audit
&& \
  shasum -a 256 -c MANIFEST.sha256)
```

Expected local sequence

Freeze study case → reproduce original result →
research evidence → stress-test analysis →
verify and adjudicate → assemble final audit.

The package is scaffolded for local import but has not passed its clean import and full end-to-end run gate. The selected ClaimBounty attempt failed qualification before scientific execution.

CHOOSE ONE SUPPORTED CLIENT ON MACOS

Install Codex or Claude Code before the local run.

The local scientific workflow requires Toone plus one supported coding-agent client. Codex and Claude Code are alternative client choices; install and authenticate one before importing and running the packaged workflow.

OFFICIAL INSTALL COMMAND SUPPLIED FOR THIS GUIDE

```
curl -fsSL https://chatgpt.com/codex/install.sh | sh
```

```
codex
```

Run `codex`, then choose **Sign in with ChatGPT**. Install and sign-in details can change, so use the current official [Codex CLI documentation](#) if the prompt differs.

Claude Code alternative

Follow Anthropic's [official Claude Code setup guide](#), then complete its documented sign-in flow. This guide does not substitute an inferred or third-party installation command.

Separate Codex review use

Codex can also inspect this repository and run contributor checks from a terminal. That review path is separate from its role as one supported client for the local Toone workflow.

Repository inspection targets

PATH	REVIEW QUESTION
contracts/openapi.yaml	Does the HTTP contract preserve identity, upload, authorization, export, and digest boundaries?
workflow/claimbounty-scientific-audit/	Are the parent routine, six child routines, resources, runner, and manifests complete?
submission/evidence/	Do public claims resolve to machine-readable status, timing, and usage records?
docs/	Do setup, architecture, benchmark, disclosure, and limitation statements agree?

A terminal assistant can read or run checks only with the permissions granted on the reviewer's machine. Do not expose private evidence, credentials, hidden answers, or customer files to any provider without authorization.

REPOSITORY ROOT

Run only the gate that answers the review question.

GOAL	COMMAND	EXPECTED OUTPUT
Verify public package integrity	<code>node scripts/generate-public-manifests.mjs --check</code>	Exit 0; reviewer, evidence, and workflow manifests match payload files.
Check public links	<code>scripts/check-public-links.mjs</code>	Exit 0; repository-relative documentation links resolve.
Run public release policy	<code>scripts/check-public-release</code>	Exit 0; no prohibited private paths, risky payloads, disclosure drift, or manifest drift.
Fast contributor loop	<code>make check-fast</code>	Contract, architecture, backend, frontend, and review-hook checks pass.
Public package gate	<code>make public-release</code>	Contract and architecture registries plus public projection policy pass.
End-to-end hosted behavior	<code>make test-system</code>	Docker profile becomes healthy; browser exercises web, email, API, storage, scanner, worker, and export verifier.
Full contributor gate	<code>make check-ci</code>	Local checks, integration, vulnerability, container build, and system suite pass.

Exact local prerequisites

- Node 22; package engine minimum 20.19.0
- pnpm 9.15.0
- Go 1.26.6 for contributor gates
- Docker Engine and Compose v2 for hosted and system paths
- Python 3.11+ for the packaged local runner

Expected repository artifacts

- `submission/reviewer/manifest.json`
- `submission/evidence/*.json`
- `workflow/claimbounty-scientific-audit/MANIFEST.sha256`
- `verified-exports/claimbounty-export/` after offline export verification

Full gates may download containers, consume substantial disk, and take longer. A visual review does not require them. Production deployment is outside this guide's verified scope.

WHAT A COMPLETE RUN SHOULD DELIVER

A decision package with evidence attached.

The final package must let a reviewer distinguish the reported result, reproduced result, defensible alternatives, unresolved conflicts, and the record of how each conclusion was reached.

Verdict

The bounded conclusion for the target claim, with status, confidence limits, and the exact evidence that supports it.

Robustness summary

How the conclusion changes across defensible data, model, covariate, and inference choices, including sensitivity limits.

Consistency findings

Agreement or conflict across paper, data, code, figures, tables, citations, and reported values.

Prioritized corrections

Actionable changes ordered by consequence, evidence, and effort, without rewriting the author's scientific judgment.

Reproduction evidence

Commands, environments, exit states, logs, numeric comparisons, tolerances, and artifact hashes needed to inspect or rerun the result.

Provenance

Source identifiers, retrieval facts, file digests, routine revisions, model usage, reviewer actions, and adjudication history.

Quality controls

- Separate technical repair from scientific comparison status.
- Preserve negative, partial, and blocked outcomes.
- Resolve public claims to released evidence.
- Keep hidden answer keys outside routines and public packages.

Review controls

- Independent method and source checks.
- Blind review where the protocol requires it.
- Adjudication before final publication.
- Evidence sealing before reference comparison.

This list is the target package contract. The protocol-deviating frozen-case attempt did not produce or pass each item.

EVIDENCE CLASSIFICATIONS MATTER

Four facts that cannot be combined.

6h49m06s

Observed internal run

Measured wall time for a prior full scientific workflow run. It is not the selected frozen benchmark case and does not establish quality.

56h

Qualified-human estimate

Estimated active work for a base case, with a 21–124 hour scenario range. It is not observed human performance and excludes unsupported assumptions about blocked time.

785.540s

One-shot ChatGPT comparator

Terminal output was truncated or malformed after two progress summaries. No required reviewer report was produced. Alpha scored 1/100, Beta 2/100, and adjudication set 1/100 with no cap and decision invalid incomplete.

Single-case boundary. The challenge recommends 10 or more cases when feasible, but this deadline-constrained evaluation contains one frozen Heliconius case. The multi-hour workflow plus case preparation, licence review, freeze validation, and isolated execution prevented a defensible 10-case evaluation. No general performance, human-quality superiority, speedup, or qualified quality comparison is published.

Selected public candidate

Heliconius Experiment 1 from Moura et al. (2023), article DOI [10.1016/j.cub.2023.06.009](https://doi.org/10.1016/j.cub.2023.06.009) and data and code record DOI [10.5281/zenodo.7985236](https://doi.org/10.5281/zenodo.7985236). The freeze is complete and validated, and the comparator is complete. The ClaimBounty exploratory attempt elapsed 3799.936556 seconds but failed exact prompt identity and separately injected two case-specific values from outside the frozen participant bundle. Those values were absent from the frozen prompt and every participant file. This is out-of-contract contamination, and the public guide omits the values. Reproduction was not executable, no scientific command ran, and G2, G3, and G4 are blocked. The attempt remains exploratory and unscored.

EVIDENCE	CLASSIFICATION	ALLOWED STATEMENT
Prior ClaimBounty run	Observed, different case	Operational timing context only
Human work	Estimated range	Planning comparator only
ChatGPT attempt	Observed terminal failure	Preserved comparator outcome
ClaimBounty attempt	Two qualification deviations, unscored	No head-to-head result or speedup

PLANNING ESTIMATE, NOT OBSERVED PERFORMANCE

Estimated active person-hours for one qualified researcher.

Eight additive work packages produce the 21-hour low, 56-hour base, and 124-hour high scenarios. The totals are an auditable estimate for the frozen scope, not a measured human benchmark, quality score, monetary cost, or speedup result.

WORK PACKAGE	TASK	LOW	BASE	HIGH
Claim map	Read manuscript and map target claim	3	5	8
Inventory	Inventory evidence and inspect data and code	2	4	8
Reproduce	Rebuild environment and reproduce result	3	8	18
Robustness design	Design robustness and sensitivity plan	2	6	12
Robustness execution	Execute and interpret robustness suite	4	14	32
Verify	Verify sources and cross-artifact consistency	2	6	14
Adjudicate	Adjudicate discrepancies and draft corrections	2	5	12
Report	Produce report, evidence table, provenance, and quality control	3	8	20
Total active person-hours		21	56	124

Main exclusions

- Unattended computation.
- Scheduler waits and blocked time.
- Data-access delays, procurement, and institutional approvals.
- Time contributed by additional specialists.

Method and source records

docs/HUMAN_BASELINE.md documents
scenario assumptions, overlap rules, sensitivity
drivers, and the full cited basis.

Machine record:
submission/evidence/human-
estimate.json.

Cited basis includes [Aczel et al. \(2021\)](#) on peer-review time, [Fišar et al. \(2024\)](#) on reproduction, and [Krafczyk et al. \(2021\)](#) on reproduction packages. These sources anchor task complexity; they do not validate the scenario totals.

VERIFIED SESSION ACCOUNTING

181,802,664 tokens; monetary cost unmeasured.

The audited records cover 109 unique CLI sessions on provider OpenAI using model gpt-5.6-sol. Retries and correction rounds are included.

RUN	SESSIONS	INPUT	CACHED INPUT	OUTPUT	TOTAL
Primary	91	157,669,522	148,934,912	1,697,629	159,367,151
Secondary	18	22,184,313	20,737,024	251,200	22,435,513
Combined	109	179,853,835	169,671,936	1,948,829	181,802,664

Counting rules

- Count each unique CLI session once.
- Cached input is already included within input.
- Reasoning output is already included within output.
- Do not add those component values to totals again.
- Repeated cumulative events and later assessment sessions are excluded.

Cost boundary

Monetary cost is unmeasured. The records identify a Pro plan but contain no contemporaneous subscription allocation, invoice, API rate, or per-run charge.

Do not apply current API prices retroactively and do not report a zero-dollar cost.

Usage evidence is not benchmark quality evidence. The 181,802,664 combined tokens span the primary and secondary included runs described in the public disclosure.

Machine-readable authority: submission/evidence/usage-and-cost.json. Narrative authority: docs/USAGE_AND_COST.md.

BEFORE ANY FINAL CLAIM

What remains open.

Scientific and product limits

- The comparator is adjudicated 1/100 and invalid incomplete.
- The evidence set contains one frozen case, not the recommended 10 or more.
- The ClaimBounty attempt has two qualification deviations and remains unscored.
- The workflow package needs a clean local import and full-run check.
- The micro1/ClaimBounty import needs release verification.
- Study cases may require extra runtimes, licensed data, or network access.
- Scanning reduces known-file risk but cannot make research code safe.
- The hosted P0 ends at export and does not adjudicate or publish.

Release gates

- ☐ Keep the protocol-deviating attempt out of comparative scoring.
- ☐ Require exact inputs and a clean comparison routine for any future scored run.
- ☐ Pass a clean-clone setup, import, and full-run test.
- ☐ Complete public evidence and redistribution review.
- ☐ Confirm deployment, repository, and competition metadata.
- ☐ Confirm fox-and-loupe account-generation provenance.
- ☐ Approve final preexisting-work and research-rights wording.
- ☐ Complete final release review.

Licensing and preexisting work

Repository software is offered under the MIT License, copyright 2026 Matheus Paranhos. Third-party dependencies and container images retain their upstream terms. Benchmark data is not redistributed in this projection and requires its own rights review before release.

The repository began from a general Go and React scaffold. Toone and its local runtime are external. ClaimBounty contracts, product surfaces, routines, package projection, and submission documents belong to this workstream, subject to Git history and competition rules. The final cutoff and classification require owner approval.

Public links

[Toone download](#) / native macOS workflow application

[Official Codex CLI documentation](#)

[Official Claude Code setup documentation](#)

[Selected benchmark data and code record](#)

Keep this guide Draft / In-Development until a qualified rerun, clean-clone test, evidence clearance, and final release review are complete.